

Deep Learning for Time Series Classification and Regression with `aeon`

Ali Ismail-Fawaz¹, Maxime Devanne¹, Antoine Guillaume²,
Anthony Bagnall³, and Germain Forestier^{1,4}

¹ IRIMAS, Université de Haute-Alsace, France
`first-name.last-name@uha.fr`

² Novahé, France
`antoine.guillaume@novahe.fr`

³ School of Electronics and Computer Science,
University of Southampton, United Kingdom
`A.J.Bagnall@soton.ac.uk`

⁴ DSAI, Monash University, Australia
`germain.forestier@monash.edu`

Abstract. We present a tutorial on deep learning for time series classification and regression with `aeon`, a Python library dedicated to time series machine learning. The tutorial first introduces the `aeon` toolkit and its unified interface, before focusing on neural architectures tailored to sequential data. We describe how participants will train and evaluate deep learning models for both classification and regression tasks using the `scikit-learn`-compatible API. In addition, we highlight a practical application in skeleton-based rehabilitation motion assessment, where deep models are applied to multivariate motion sequences from the Rehab-Pile archive. This application demonstrates how learned representations can support real-world healthcare scenarios involving movement analysis and assessment. More details about the tutorial schedule and materials are available on the tutorial webpage:

<https://aeon-tutorials.github.io/ADBIS-2026/>

Keywords: Time series · Deep learning · Classification · Regression · Rehabilitation · `aeon`

1 Introduction

Time series data [15] appears in many scientific and industrial contexts, ranging from biomedical signals to sensor recordings and financial measurements. While traditional machine learning approaches [5] remain relevant, deep learning methods [12,16] have gained significant traction due to their ability to automatically learn hierarchical representations directly from raw sequences.

Time series classification (TSC) [15] assigns categorical labels to sequences, whereas time series regression (TSER) [7] predicts continuous outcomes. Both tasks benefit from neural architectures such as convolutional neural networks

(CNNs), residual networks, and hybrid approaches designed to capture temporal dependencies and spatial structure.

This tutorial focuses on deep learning models implemented in **aeon**, and concludes with a domain-specific application to human motion rehabilitation.

2 Related Work on Deep Learning for Time Series Classification and Regression

Several deep learning architectures have been successfully applied to both Time Series Classification (TSC) and Time Series Regression (TSER). Fully Convolutional Networks (FCN) employ stacked convolutional layers with global average pooling to learn discriminative temporal features [19]. ResNet introduces residual connections to facilitate the training of deeper convolutional networks [19]. InceptionTime and its hierarchical extension H-InceptionTime use multi-scale convolutional modules to capture temporal patterns at different receptive fields [13,11]. LITETime and LITEMVTime are lightweight Inception-based models designed for efficient univariate and multivariate time series learning [8,10]. ConvTran combines convolutional feature extraction with transformer-based self-attention [6], whereas VanTran relies solely on a vanilla transformer encoder to model long-range temporal dependencies [18]. These architectures have demonstrated strong performance across both TSC and TSER tasks.

3 Overview of the **aeon** Toolkit

aeon [14] is an open-source Python library designed for time series machine learning. It provides a unified interface for a wide range of learning tasks on sequential data, including classification, regression, clustering, forecasting, anomaly detection, segmentation, and similarity search. The toolkit follows the design principles of **scikit-learn** [17], exposing a consistent estimator API based on the **fit** and **predict** paradigm, which allows integration with existing machine learning workflows.

A key objective of **aeon** is to make state-of-the-art time series algorithms accessible through a common framework while supporting reproducibility and practical usability. The library supports both univariate and multivariate time series and includes utilities for data loading, preprocessing, evaluation, and experimentation. In this tutorial, we focus specifically on its deep learning functionality for classification and regression, showing how neural models can be used within the same interface as other **aeon** estimators.

4 Deep Learning in **aeon**

The **aeon** toolkit provides implementations of modern neural architectures for time series learning, following a consistent estimator interface compatible with

`scikit-learn`. This design allows users to train deep models using familiar commands while retaining access to time series specific data handling and evaluation utilities.

The deep learning module includes several neural architectures that have been adapted for time series data. These include convolutional models such as Fully Convolutional Networks (FCN), which learn local temporal patterns, and Residual Networks (ResNet) [19], which enable the training of deeper models through skip connections. More recent architectures, such as Inception-based models [13,8,10], capture patterns at multiple temporal scales, while hybrid approaches [11,1] combine different architectural components to improve representation learning.

These models support multivariate sequences [2] and can leverage hardware acceleration when available. They are particularly useful when the goal is to learn representations directly from raw or lightly processed time series, reducing the need for handcrafted features.

5 Using Deep Learning Models

The practical part of the tutorial demonstrates how deep learning models can be trained using the same workflow as other estimators in `aeon`. Participants will see how to load benchmark datasets [4], instantiate a neural model, fit it to training data, and generate predictions on unseen cases.

5.1 Classification

For classification, the objective is to assign each input time series to one of a finite number of classes. The following example loads a classification dataset, trains a Fully Convolutional Network, and produces class probability estimates for the test set.

```

1 from aeon.datasets import load_classification
2 from aeon.classification.deep_learning import FCNClassifier
3
4 X, y = load_classification("ArrowHead", split="TRAIN")
5 X_test, _ = load_classification("ArrowHead", split="TEST")
6
7 clf = FCNClassifier(n_epochs=50, batch_size=16)
8 clf.fit(X, y)
9
10 probs = clf.predict_proba(X_test)

```

5.2 Regression

For regression, the target variable is continuous rather than categorical. The workflow remains similar, but the model outputs numerical predictions rather than class labels or class probabilities.

```

1 from aeon.datasets import load_regression
2 from aeon.regression.deep_learning import ResNetRegressor
3
4 X, y = load_regression("BarCrawl6min", split="TRAIN")
5 X_test, _ = load_regression("BarCrawl6min", split="TEST")
6
7 reg = ResNetRegressor(n_epochs=50, batch_size=16)
8 reg.fit(X, y)
9
10 preds = reg.predict(X_test)

```

6 Tutorial Content Description

The tutorial is structured to guide participants progressively from general concepts to practical implementation. It begins with an introduction to time series machine learning and the role of deep learning methods in this context. We then present the `aeon` toolkit, highlighting its design principles and how it simplifies the use of advanced algorithms for sequential data.

The core of the tutorial focuses on deep learning models for time series classification and regression. We explain the intuition behind common neural architectures and discuss their strengths for modelling temporal data. Participants will learn how convolutional, residual, and multi-scale architectures can be used to learn discriminative representations from univariate and multivariate time series.

This is followed by a coding demonstration in which participants will learn how to load datasets, configure models, train estimators, and evaluate predictions using `aeon`. The tutorial will emphasize practical usage rather than implementation details of neural networks from scratch, making the session accessible to participants who want to apply these methods in their own work.

In the final part, we present a domain-specific application on skeleton-based human motion rehabilitation using the Rehab-Pile archive [9]. This example illustrates how the presented methods can be applied to real-world motion data and highlights the challenges and opportunities associated with rehabilitation-oriented time series analysis.

7 Tutorial Format and Schedule

The tutorial is designed as a 90-minute session combining conceptual presentation and practical demonstration. The first part, lasting approximately one hour, introduces the fundamental concepts of time series deep learning and the `aeon` framework. This includes explanations of neural architectures and their application to classification and regression tasks.

The second part consists of a hands-on coding demonstration of around 20 minutes, where participants will follow the implementation of deep learning models using `aeon`. This segment focuses on practical usage, including data loading, model training, and prediction.

The session concludes with a 10-minute question-and-answer segment, allowing participants to discuss specific use cases and clarify technical details.

8 Target Audience and Prerequisites

This tutorial is intended for researchers and practitioners working in data science and machine learning who are interested in temporal or sequential data. It is particularly relevant for those who wish to apply deep learning techniques to time series problems in domains such as healthcare, sensor analysis, or human motion understanding.

Participants are expected to have a basic understanding of machine learning concepts and familiarity with Python programming. Prior exposure to deep learning or time series analysis can be beneficial, but is not strictly required, as the tutorial will introduce the necessary concepts before moving to practical examples.

The proposed tutorial builds on a series of previous tutorials and demonstrations delivered by the authors on time series machine learning and deep learning methods:

- Tutorial on deep learning for time series classification with aeon at EGC 2024 ⁵
- Tutorial on time series machine learning with aeon at ECML 2024 ⁶
- Tutorial on time series classification and regression with aeon at KDD 2024 [3] ⁷
- Tutorial on imperfect data time series classification with aeon at DSAA 2025 ⁸
- Demonstration at ECML 2025 on time series classification and regression with aeon ⁹

These previous events have allowed us to refine both the pedagogical structure and the practical content of the tutorial, ensuring a balanced combination of theoretical background and hands-on experience.

9 Application: Skeleton-Based Rehabilitation with Rehab-Pile

Beyond benchmark datasets, deep learning models for time series are particularly suited to structured motion data. One important application domain is rehabilitation, where patient movements are recorded as sequences of joint coordinates over time.

⁵ <https://germain-forestier.info/e-egc.pdf>

⁶ <https://aeon-tutorials.github.io/ECML-2024/>

⁷ <https://aeon-tutorials.github.io/KDD-2024/>

⁸ <https://aeon-tutorials.github.io/DSAA-2025/>

⁹ <https://aeon-tutorials.github.io/ECML-Demo-2025/>

Skeleton-based motion data [9] can be represented as multivariate time series, with each dimension corresponding to joint positions or angles. These sequences capture both spatial configuration and temporal evolution, making them well suited to deep neural models. In this setting, the objective may be to recognize exercise types, assess execution quality, detect compensatory movements, or predict continuous rehabilitation scores.

In this tutorial, we consider the Rehab-Pile archive, a collection of rehabilitation exercise recordings designed for movement assessment. The application will show how skeleton sequences can be formatted as time series data and used with deep learning models in `aeon`. This part of the tutorial connects the general methodology to a concrete healthcare scenario.

10 Tutors’ Short Biographies

10.1 In-person Tutorial Presenters

Ali Ismail-Fawaz (AF) is an Associate Professor in Computer Science at the University of Haute-Alsace since 2025. He received his PhD in Computer Science from the University of Haute-Alsace in 2025. After completing his doctorate, he joined the University of Haute-Alsace in Mulhouse, France, as an Associate Professor, where he is part of the IRIMAS research institute. His main research topic is focused on time series machine learning, while addressing different applications mainly skeleton-based human motion and EEG signals. He is a core developer of the `aeon` toolkit and is leading the deep learning part of the package with a recent kick off for a deep forecasting module and is partner in the P16 Inria grant for the development of *aeon*, that is supporting this tutorial.¹⁰ His work has addressed different tasks such as classification, regression, generation, and clustering (<https://hadifawaz1999.github.io/>).

Germain Forestier (GF) is a university professor at the University of Haute-Alsace in Mulhouse France. He is part of the IRIMAS research institute, for which he is head of the computer science department. His research is mainly oriented around time series analysis, with a speciality in deep learning models (<https://germain-forestier.info/>). He is co-author of the deep learning review for time series classification for which the deep learning portions of the tutorial will be based on mostly, as well as new papers which he is co-authored as well. His contributions to the deep learning community in time series classification have been significant recently. He is also a core developer of the `aeon` toolkit and PI of the P16 Inria grant for the development of *aeon*.

Antoine Guillaume (AG) is an AI engineer at Novahé. He is currently working in the industry to create, maintain, and deploy software products focused on generative AI, agents, and time series. He is also engaged in the study of time series similarity search within the context of rare event prediction for critical industrial systems and shapelet-based methods. He is a core developer of the `aeon` toolkit. (<https://baraline.github.io/>).

¹⁰ <https://p16.inria.fr/fr/>

10.2 Tutorial Contributors

Anthony Bagnall (AB) is a Professor in Computer Science at the University of Southampton. His primary research interest is in developing algorithms for time series machine learning, including classification, regression and clustering, and finding novel areas of application. He has ongoing research collaborations with scientists on four continents and a track record of consultancy with industrial partners. He is an action editor for ECML 2026, is on the editorial board for the Data Mining and Knowledge Discovery journal and regularly reviews for data mining conferences such as KDD, ICDM and SDM. He helps maintain the time series classification archive (<https://www.timeseriesclassification.com>), including the UCR archive, that has been used in thousands of publications. More details are available at: <https://www.southampton.ac.uk/people/62tpqc/professor-tony-bagnall>.

Maxime Devanne (MD) is an Associate Professor in Computer Science at the University of Haute-Alsace since 2019. He received his PhD in Computer Science from both the University of Lille (France) and the Università di Firenze (Italy) in 2015. From 2016 to 2019, he was a postdoctoral researcher at IMT Atlantique and the Hospital of Brest. His main research interests are time series and human motion analysis with deep learning-based approaches (<https://maxime-devanne.com>).

11 Conclusion

We presented a tutorial on the deep learning capabilities of the `aeon` toolkit for time series classification and regression, along with an application to skeleton-based rehabilitation. By combining flexible neural architectures with a unified interface, `aeon` provides a practical framework for both research and real-world deployment. The Rehab-Pile example highlights the potential of deep time series models in healthcare settings, where accurate and automated analysis of movement data is increasingly important.

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References

1. Badi, O., Devanne, M., Ismail-Fawaz, A., Abdullayev, J., Lemaire, V., Berretti, S., Weber, J., Forestier, G.: Cocalite: A hybrid model combining catch22 and lite for time series classification. In: 2024 IEEE International Conference on Big Data (BigData). pp. 1229–1236. IEEE (2024)
2. Bagnall, A., Dau, H.A., Lines, J., Flynn, M., Large, J., Bostrom, A., Southam, P., Keogh, E.: The uea multivariate time series classification archive, 2018. arXiv preprint arXiv:1811.00075 (2018)

3. Bagnall, A., Middlehurst, M., Forestier, G., Ismail-Fawaz, A., Guillaume, A., Guijo-Rubio, D., Tan, C.W., Dempster, A., Webb, G.I.: A hands-on introduction to time series classification and regression. In: Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. pp. 6410–6411 (2024)
4. Dau, H.A., Bagnall, A., Kamgar, K., Yeh, C.C.M., Zhu, Y., Gharghabi, S., Ratanamahatana, C.A., Keogh, E.: The ucr time series archive. *IEEE/CAA Journal of Automatica Sinica* **6**(6), 1293–1305 (2019)
5. Dempster, A., Petitjean, F., Webb, G.I.: Rocket: exceptionally fast and accurate time series classification using random convolutional kernels. *arXiv preprint arXiv:1910.13051* (2019)
6. Foumani, N.M., Tan, C.W., Webb, G.I., Salehi, M.: Improving position encoding of transformers for multivariate time series classification. *Data Mining and Knowledge Discovery* (2024)
7. Guijo-Rubio, D., Middlehurst, M., Arcencio, G., Silva, D.F., Bagnall, A.: Unsupervised feature based algorithms for time series extrinsic regression. *Data Mining and Knowledge Discovery* (2023)
8. Ismail-Fawaz, A., Devanne, M., Berretti, S., Weber, J., Forestier, G.: Lite: Light inception with boosting techniques for time series classification. In: 2023 IEEE 10th International Conference on Data Science and Advanced Analytics (DSAA). pp. 1–10. IEEE (2023)
9. Ismail-Fawaz, A., Devanne, M., Berretti, S., Weber, J., Forestier, G.: Deep learning for skeleton based human motion rehabilitation assessment: A benchmark. *arXiv preprint arXiv:2507.21018* (2025)
10. Ismail-Fawaz, A., Devanne, M., Berretti, S., Weber, J., Forestier, G.: Look into the lite in deep learning for time series classification. *International Journal of Data Science and Analytics* **20**(4), 4029–4049 (2025)
11. Ismail-Fawaz, A., Devanne, M., Weber, J., Forestier, G.: Deep learning for time series classification using new hand-crafted convolution filters. In: 2022 IEEE International conference on big data (Big Data). pp. 972–981. IEEE (2022)
12. Ismail Fawaz, H., Forestier, G., Weber, J., Idoumghar, L., Muller, P.A.: Deep learning for time series classification: a review. *Data mining and knowledge discovery* **33**(4), 917–963 (2019)
13. Ismail Fawaz, H., Lucas, B., Forestier, G., Pelletier, C., Schmidt, D.F., Weber, J., Webb, G.I., Idoumghar, L., Muller, P.A., Petitjean, F.: Inceptiontime: Finding alexnet for time series classification. *Data mining and knowledge discovery* **34**(6), 1936–1962 (2020)
14. Middlehurst, M., Ismail-Fawaz, A., Guillaume, A., Holder, C., Guijo-Rubio, D., Bulatova, G., Tsaprounis, L., Mentel, L., Walter, M., Schäfer, P., et al.: aeon: a python toolkit for learning from time series. *Journal of Machine Learning Research* **25**(289), 1–10 (2024)
15. Middlehurst, M., Schäfer, P., Bagnall, A.: Bake off redux: a review and experimental evaluation of recent time series classification algorithms: M. middlehurst et al. *Data Mining and Knowledge Discovery* **38**(4), 1958–2031 (2024)
16. Mohammadi Foumani, N., Miller, L., Tan, C.W., Webb, G.I., Forestier, G., Salehi, M.: Deep learning for time series classification and extrinsic regression: A current survey. *ACM Computing Surveys* **56**(9), 1–45 (2024)
17. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., et al.: Scikit-learn: Machine learning in python. *the Journal of machine Learning research* **12**, 2825–2830 (2011)

18. Petrovich, M., Black, M.J., Varol, G.: Action-conditioned 3d human motion synthesis with transformer vae. In: Proceedings of the IEEE/CVF international conference on computer vision. pp. 10985–10995 (2021)
19. Wang, Z., Yan, W., Oates, T.: Time series classification from scratch with deep neural networks: A strong baseline. In: 2017 International joint conference on neural networks (IJCNN). pp. 1578–1585. IEEE (2017)